

Pediatric Cardiac Arrest – General

This protocol should be followed for all pediatric cardiac arrests.

- If an arrest is of a known traumatic origin refer to the **Traumatic Arrest-Treatment Protocol**.
- If it is unknown whether the arrest is traumatic or medical, and the patient does not meet dead on scene criteria per **Dead on Scene Termination of Resuscitation-Procedure Protocol**, start CPR and continue with this protocol.
- If patient is hypothermic refer to **Hypothermic/Frostbite-Treatment Protocol** for warming techniques when applicable.

Note: Primary cardiac arrest in the pediatric patient is rare. Most arrests are secondary to respiratory failure. Maintaining basic airway management techniques unless unable or ineffective. Advanced airway insertion attempts should be performed only if BLS airway management is ineffective. Keep CPR interruptions to a minimum. Medications given during cardiac arrest are given IV or IO.

HIGH QUALITY CPR & DEFIBRILLATION

- CPR and electrical therapy should be consistent with current American Heart Association guidelines. For all patients, **anterior/posterior placement** of pads is preferred and should be used, if possible, and if defibrillation not delayed.
- Once arrest is confirmed, emphasis should be on avoiding interruptions in CPR.
- CPR should be done in accordance with current guidelines established by the American Heart Association.
- Compressions at least 1.5" in depth for infants, 2" in depth for children (at least one third the anteroposterior diameter of the chest).
- Compression rate of at least 100-120 per minute
- Allow full chest recoil with each compression for maximum perfusion.
- Avoid excessive ventilation (volume and rate).
- Continue CPR with minimal interruptions, changing the rescuer doing compressions
- Verify CPR quality frequently and any time rescuer providing compressions or ventilations change.
- Change rescuer performing compressions at least every 2 minutes to avoid fatigue.
- Interruption in compressions must be less than 10 seconds
- If an advanced airway is placed, provide continuous CPR, without pauses for ventilation and ventilate at 20 breaths per minute or 1 breath every 3 seconds

OPERATIONAL CONSIDERATIONS

1. Prior to advanced airway placement, utilize ventilation periods to visualize the ECG rhythm without compression artifact, this will allow you to plan for the assessment period at the end of the two-minute CPR cycle.
2. If AED has been applied by BLS personnel, skip to appropriate place in protocol that

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incorporates previous care. ALS personnel should switch to manual defibrillator after initial AED defibrillation or place AED in manual mode.

PROCEDURE

1. Request additional assistance, as needed, and initiate ALS response, if available.
2. Confirm Arrest
 - a. Assess for signs of normal breathing. Agonal breathing is associated with cardiac arrest.
 - b. Check a carotid or brachial pulse as age appropriate for no more than 10 seconds.
3. Initiate CPR or continue CPR if already in progress and apply and use AED/manual defibrillator per **Electrical Therapy-Procedure Protocol** as soon as possible. Use AED pediatric pads and settings per AED manufacturer instructions for use.
4. Ensure CPR quality
 - a. Manual chest compressions remain the standard of care. Mechanical chest compression devices may be a reasonable alternative to conventional CPR in specific settings where the delivery of high quality manual compression may be challenging or dangerous for the provider (e.g., limited rescuers, prolonged CPR, CPR during hypothermic cardiac arrest, CPR in a moving ambulance). An FDA approved, MCA authorized mechanical CPR device operating at the manufacturer's pre-set rate may be utilized. See **Mechanical Chest Compression Device-Procedure Protocol** for age/weight requirements and limitations. (MCA Optional)
 - b. An impedance threshold device may be utilized during CPR for children > 10kg (if available). Device should be discontinued immediately upon return of spontaneous circulation. See **Impedance Threshold Device-Procedure Protocol** (MCA Optional Protocol)
5. Establish a patent airway, maintaining C-Spine precautions if indicated, beginning with BLS airway adjuncts and a BVM with high flow oxygen. Ventilations with BVM (2-rescuer technique) and airway adjuncts are at least as effective as endotracheal intubation in children.
 - a. 2-person bag-valve-mask ventilation with oral airway should be standard technique
 - b. If only 2 rescuers, rescuer performing compressions can squeeze bag while 2nd rescuer maintains face to mask seal with both hands
 - c. If unable to ventilate or unable to maintain a patent airway, establish an advanced airway per the **Airway Management-Procedure Protocol**. (Supraglottic airways are first choice advanced airway for pediatrics when age approved sizes are available)
 - i. All advanced airways (includes supraglottic) require EtCO₂ monitoring.
-   6. If Return of Spontaneous Circulation (ROSC) has **not** been achieved after three, two-minute cycles of CPR AND ALS is not available or delayed, contact Medical Control to discuss initiation of BLS transport while continuing to focus on high quality CPR.
7. Reassess ABC's as indicated by rhythm or patient condition change. Pulse checks should take no more than 10 seconds. If no pulse after 10 seconds, assume pulselessness, continue CPR beginning with compressions.

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8. Continuously monitor EtCO₂ per MCA selection in End-Tidal Carbon Dioxide Monitoring-Procedure Protocol.
 - a. EtCO₂ of 0 is indicative of failed airway.
 - b. If EtCO₂ is <10 mmHG, attempt to improve CPR quality. If CPR quality good, may indicate futility state.
 - c. Monitor EtCO₂ for trends and indications of patient status.
9. Start an IV/IO **NS** or **LR** KVO. IO may be the first choice. See **Vascular Access & IV Fluid Therapy-Procedure Protocol**.
10. Check rhythm, every 2 minutes, defibrillate according to MI MEDIC card. If MI MEDIC are not available:
 - a. Initial defibrillation at 2 J/kg (or closest energy setting specific to defibrillator being utilized), and continue CPR.
 - b. Subsequent defibrillations must be at least 4 J/kg, but may escalate to 10J/kg or adult dosage.
11. Administer **epinephrine** according to MI MEDIC cards.
 - a. Initial dose should ideally be administered within 5 minutes of ALS/LALS contact of confirmed pediatric cardiac arrest.
 - b. If MI MEDIC cards are not available administer:
 - i. 1 mg/10 ml, 0.01 mg/kg (0.1 ml/kg)
 - ii. Max dose 1mg (10 ml)
 - iii. Repeat every 3-5 minutes
12. If shockable rhythm persists administer antiarrhythmic (per MCA selection) according to MI MEDIC cards.
 - a. If MI MEDIC cards are not available administer antiarrhythmic (per MCA selection) as follows:

Per MCA Selection

- ☐ **Amiodarone** 5 mg/kg (max single dose 300 mg) IV/IO (May repeat twice) Do not exceed 450 mg total IV/IO
- or
- ☒ **Lidocaine** 1 mg/kg IV/IO (May repeat 0.5 mg/kg twice at 5-10 minute intervals. Maximum 3 doses total)

13. Identify and treat reversible causes of arrest
 - a. Hypovolemia (including vomiting/diarrhea)– Administer 20 ml/kg **NS** or **LR** IV/IO bolus
 - b. Hypoglycemia – check blood glucose (may be MFR skill, see **Blood Glucose Testing-Procedure Protocol**)
 - i. If blood glucose is less than 60 mg/dL administer dextrose according to MI MEDIC cards.
 - ii. If MI-MEDIC unavailable, administer **dextrose** 0.5 g/kg per Pediatric Altered Mental Status.
 - c. Tension pneumothorax – see **Pleural Decompression-Procedure Protocol**
 - d. Hyperkalemia (renal failure) – Contact Medical Control
 - i. Administer **calcium chloride 10%** per MI MEDIC cards
 1. If MI MEDIC cards are unavailable administer 20 mg/kg (0.2 ml/kg), max single dose 1 gm

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- ii. FLUSH line with 20 mL **NS** between calcium chloride and sodium bicarbonate administration.
- iii. Administer **sodium bicarbonate** per MI MEDIC cards
 2. If MI MEDIC cards are unavailable administer 1 mEq/kg IV/IO
5. If ROSC is not achieved, continue resuscitation while contacting Medical Control
 -  a. **BLS/LALS:** If ROSC has not been achieved and ALS is not available or is delayed, contact Medical Control after **20 minutes** of high-quality CPR for further direction AND before initiating transport. Continue high quality CPR unless directed otherwise by Medical Control per **Dead on Scene & Termination of Resuscitation Protocol**.
 -  b. **ALS:** If ROSC is not present after **30 minutes of ALS time** contact Medical Control for further direction AND before initiating transport.
 - c. Continue high quality CPR unless directed otherwise by Medical Control per **Dead on Scene & Termination of Resuscitation Protocol**.

Notes:

1. Chest Compression Fraction (CCF) is the proportion of time during cardiac arrest when compressions are being performed. CCF should be as high as possible: ideally greater than 80% (AHA, ACLS, pg.115)
-  2. Identify and communicate to Medical Control potentially reversible causes. Treat EMS reversible causes, using other protocols, as applicable.
 - A. Hyper/hypokalemia (known renal failure), other metabolic disorders
 - B. Hypothermia
 - C. Hypovolemia (including vomiting/diarrhea)
 - D. Hypoxia
 - E. Hydrogen ion excess (acidosis)
 - F. Toxins/ overdose (e.g., beta-blocker or calcium channel-blocker)
 - G. Tamponade
 - H. Tension pneumothorax
 - I. Thrombosis (pulmonary or coronary)
3. Routine use of **sodium bicarbonate** and **calcium chloride** in cardiac arrest is not indicated.
4. If ROSC is achieved refer to **Pediatric Return of Spontaneous Circulation - Treatment Protocol**

Medication Protocols

Amiodarone
Calcium Chloride
Dextrose
Epinephrine
Lidocaine
Sodium Bicarbonate